

Module 2

1. Consider the following relation schema:

SAILORS (sid, sname, rating, age)

RESERVES (sid, bid, day)

BOATS (bid, bname, color)

Express the following queries in relational algebra:

- a. Find the names of sailors who have reserved at least one boat.
 - b. Find the names of sailors who have reserved a red or a green boat.
 - c. Find the names of the sailors who have reserved all color boats.
2. Consider the following relational schema in which an employee can work in more than one department:

EMPLOYEE (EMP-ID, EMP-NAME, SALARY)

WORKS (EMP-ID, DEPT-ID)

DEPARTMENT (DEPT-ID, DEPT-NAME, MGR-ID, FLOOR-NO)

Express the following queries in relational algebra:

- a. Find the names of all employees who work on the 1st floor and earn less than INR 70,000.
 - b. Find the EMP-ID of all employees who do not work for a department located at 2nd floor.
 - c. Find the name of all employees who earn more than INR 35,000 and work in either the IT department or the computer science department.
3. Write the relational algebra and SQL queries for the relation given below.

Employee (name, street, city)

Works(ename,cname,salary)

Company(cname,city)

Manages(ename,mgrname)

(a) Find name and city of all employees who work for First bank corporation and salary>10000.

(b) Retrieve the names of all employees who live in the same city and on the same street as the managers.

(c) Find the names of all employees who earn more than every employee of a small bank corporation.